

FIN305 Risk Management for Business – Group Project
(Worth 30% of the Total Grade)

~~-----This is a provisional version, subject to moderation-----~~

Learning Outcomes:

- *Demonstrate knowledge and understanding of structured methods of assessing financial risks*
- *Apply techniques for capturing and managing financial risks*
- *Evaluate and summarize economic and financial events related to risk management*

Project Content

Students must select **at least 5 common stocks** from the Shanghai and Shenzhen Stock Exchanges. The sample period must cover **at least 15 years of daily data** (e.g., 2005–2021) and have at least **2,520 non-missing observations**. If stocks are chosen, ensure all the stocks come from the same industry. Data can be obtained from **Wind, Thomson Reuters, CSMAR, Bloomberg, or WRDS**.

You must carry out the following tasks in **RStudio**, document your code, and write a concise report interpreting the results.

Question 1

- a) Calculate daily log returns from **adjusted closing prices**. Report descriptive statistics (mean, median, std, skew, kurtosis, min, max, quartiles) and plot the return series.
- b) Plot the efficient frontier of the 5 assets and identify the **Global Minimum Variance (GMV)** portfolio and **Tangency (maximum Sharpe ratio)** portfolio. Present the weights for the two portfolios.
[Hint: randomly generate 10,000 possible portfolios to visualize the efficient frontier and obtain the one with maximum GMV and SR. Sharpe ratio = $(r_p - r_f) / \text{sig}_p$ where r_p is the expected portfolio return and sig_p is the standard deviation of the portfolio return]
- c) Impose realistic **constraints** (e.g., no short-selling, weights ≥ 0 , weights ≤ 0.25). Get the weights for portfolios with maximum GMV and SR again. Compare the performance between constrained vs. unconstrained portfolios.

[10+10+10=30 marks]

Question 2

Please employ the optimal portfolio (with the highest Sharpe ratio) obtained in Question 1 in addressing the following questions:

- a) Fit a copula to the joint distribution of the 5 assets. Report the estimated parameters (e.g., correlation matrix).
- b) Based on the simulated joint distribution, compute the **1-day 95% portfolio VaR** for the optimal portfolio obtained in Question 1. Compare this copula-based VaR with the variance–covariance and historical simulation VaRs. Explain the difference in VaRs between the different methods.
- c) Use the first 1000 observations to estimate the 1-day VaR at a 90% confidence level, apply a rolling-window procedure to estimate VaR for the subsequent data points and then perform backtesting to evaluate the accuracy of your VaR model based on the

Copula model. For backtesting, consider the binomial test at a 5% significance level to assess whether the number of VaR breaches is consistent with the expected frequency.

[Hint: in a rolling-window procedure, the in-sample data is always 1000 observations, and with each step, you move the window forward by one day. For instance, use days 1-1000 to calculate the VaR for day 1001, then use days 2-1001 to calculate the VaR for day 1002, and continue this process until you reach the end of the dataset.]

[10+10+20=40 marks]

Question 3

- a) Estimate a **GARCH(1,1)** model for the GMV and Tangency portfolios. Report the estimated parameters. Plot the conditional volatilities.
- b) With the results in a), discuss how volatility behaves during extreme market conditions, e.g., 2008 financial crisis and covid-19 pandemic... What do you think could lead to the volatility spikes during these periods?
- c) Using the GARCH(1,1) model for the optimal portfolio obtained in Question 1, forecast the 1-day-ahead conditional volatility. Compute the 1-day 95% VaR based on this forecast under the variance–covariance (normal) method. Compare the GARCH-based VaR with the historical simulation, variance–covariance VaRs, and the copula-based VaR from question 2.

[10+10+10=30 marks]

How to create groups

Each project group shall include 4-6 students. Students are free to choose their group members till **13 October, 2025**.

Project Output

- 1) A single PDF document¹ (the overall word limit on this file is 1,500 words maximum, excluding tables/figures/references/R code):
 - a) Provide solutions to each question: address every part of the assignment thoroughly, ensuring that all questions are answered in full.
 - b) Use of R Programming: for questions that require the use of R programming, include the **relevant R code** used for solving the problem. After running the code, present the results clearly, summarizing the outcomes. Where the results are detailed, use tables or figures to display them effectively.
 - c) Discussion of Economic Phenomena: For questions that ask for a discussion on economic implications or phenomena, provide a detailed explanation that outlines your reasoning and logic clearly. Make sure to justify your conclusions based on evidence and analysis. Ensure that you reference any additional sources of information consulted, using the Harvard referencing style.
- 2) A csv file containing the original data of the five stocks and the returns you calculated
- 3) R script containing all the code (with comments) used to generate the results

¹ A4 paper and 1.5 or double spacing for the PDF file. Pages should be numbered.

Please name all the files listed above using your group number (shown on learning mall) with the name of submitter. **Only one member**, on behalf of the group, needs to submit the assignment through the dropbox on learning mall.

Each group must fill in and sign the **group contribution form** and insert the form into the PDF file as the first page (the form is attached at the end of this document).

Table 1 shows how the overall mark of a hypothetical student is calculated for the group project.

Table 1 Student mark summary for the group project

Suppose you had a 4-member group. The group earned an 80 on the report. The default assumption is each person contributed 25%. The grades will be adjusted by the percentage above or below the presumed percentage as follows:

Individual Name:	%Contribution (X)	Grade $80+80*(X-25\%)$
Joe	25%	80
Jane	25%	80
Mike	35%	$80+80*10\%=88$
Sue	15%	$80+80*(-10\%)=72$
	100%	

Note: In case that you cannot reach a consensus, I will meet with all members of a group at one time. I hope that I will not have to do this because I am not privy to the group's actual contributions and would prefer you coming to a consensus by yourselves. The module leader reserves the rights to make special adjustments for individual marks in extreme cases.

Appendix A

GROUP MEMBER CONTRIBUTION FORM

FIN305 – Group Assignment

The purpose of this document is to provide each group with an opportunity to reward or punish an individual's contribution to the group. All members of the group should discuss this form, fill in this form as directed, sign this form, and submit it with the report as the coversheet.

We agree that all group members made a valuable contribution. Please adjust our grades based on the following percentage of contribution.

Individual Name (print):	%Contribution to the group project	Authorship contribution statement ²
	%	
	%	
	%	
	%	
	%	
	%	
	100%	

Signatures:

1. _____
2. _____
3. _____
4. _____

² This is to provide an accurate and brief description of each group member's contribution to the work. Examples are such as data collection, coding, empirical analysis, writing....